



Genome-wide analysis of Family-1 UDP-glycosyltransferases in soybean confirms their abundance and varied expression during seed development

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ABSTRACT

Family-1 UDP-glycosyltransferases (EC 2.4.1.x; UGTs) are enzymes that glycosylate aglycones into glycoside-associated compounds with improved transport and water solubility. This glycosylation mechanism is vital to plant functions, such as regulation of hormonal homeostasis, growth and development, xenobiotic detoxification, stress response, and biosynthesis of secondary metabolites. Here, we report a genome-wide analysis of soybean that identified 149 putative UGTs based on 44 conserved plant secondary product glycosyl-transferase (PSPG) motif amino acid sequences. Phylogenetic analysis against 22 referenced UGTs from *Arabidopsis* and maize clustered the putative UGTs into 15 major groups (A–O); J, K, and N were not represented, but the UGTs were distributed across all chromosomes except chromosome 04. Leucine was the most abundant amino acid across all 149 UGT peptide sequences. Two conserved introns (C₁ and C₂) were detected in the most intron-containing UGTs. Publicly available microarray data on their maximum expression in the seed developmental stage were further confirmed using Affymetrix soybean IVT array and RNA sequencing data. The UGT expression models were designed, based on reads per kilobase of gene model per million mapped read (RPKM) values confirmed their maximally varied expression at globular and early maturation stages of seed development.

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1. Introduction

Soybean (*Glycine max*) is among the world's most important crops as a major source of edible oil, vegetable protein, and active compounds such as isoflavones, saponins, phytosterols, and tocopherols. The availability of the soybean genome and substantial germplasm resources also make it a popular model plant for the study of oil traits, root nodulation, secondary metabolism, and in comparative genomics.

The glycosyltransferase gene family plays an important role in higher plants. The encoded enzymes perform glycosylation, the addition of sugar moieties from activated nucleotides to a wide range of aglycone active sites. After cyclization and oxidation, glycosylation is the final step in the biosynthesis of many secondary plant products (e.g., terpenoids, flavonoids, saponins, and steroids; Hironaka et al., 2000). The same process also modifies phytohormones and xenobiotics, thus regulating hormonal homeostasis and xenobiotic detoxification (Bowles et al., 2005; Hou et al., 2004; Jones and Vogt, 2001; Loutre et al., 2003). Following these reactions, aglycones alter in stability, hydrophilicity, localization, and chemical bonding with other macromolecules (Bowles et al., 2005; Kramer et al., 2003; Kristensen et al., 2005). Specifically, uridine-diphosphate (UDP)-glycosyltransferases (UGTs) are a major portion of prokaryotic and eukaryotic genomes. Moreover, UGT association has been explored in both Plantae and Animalia.

Proteins classified as carbohydrate-active enzymes (CAZy) probably evolved from the same ancestral sequence, exhibiting conserved reaction mechanisms and sharing homologous struc-

Abbreviations: UGTs, UDP-glycosyltransferases; PSPG, plant secondary product glycosyl-transferase; RPKM, reads per kilobase of gene model per million mapped reads; CAZy, carbohydrate-active enzymes; MEME, multiple em for motif elicitation; UDP, uridine-diphosphate; PSWM, position specific weight matrix; BLASTP, basic local alignment search tool; E-value, expectation value; Ks, synonymous substitutions; GS, globular stage; HS, heart stage; CS, cotyledonary stage; EMS, early maturation stage; MAS, affytrix algorithm.

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tural folds. Among the 99 diverse families currently in the CAZy database (www.cazy.org), UGTs roughly correspond to CAZy family 1. This categorization is based on percentage sequence similarity and substrate identity (Campbell et al., 1997; Coutinho et al., 2003). In comparison with other organisms, plant genomes are overrepresented by UGTs, which possess a highly conserved motif of 44 amino acids (the PSPG—plant secondary product glycosyltransferase—box) (Hughes and Hughes, 1994) in their C-terminus, observable via motif analysis tools (e.g., MEME). The PSPG box is suggested to be the nucleotide diphosphate sugar binding site (Vogt and Jones, 2000). As expected from the name, UGTs attach UDP-converted glucose to aglycones, resulting in activated sugars such as glucuronic acid, rhamnose, galactose, and xylose (Merken and Beecher, 2000). These multiple or single glycosylation events can occur at a variety of active groups, including —NH₂, C—C, —COOH, —OH, and —SH (Bowles et al., 2006; Breton et al., 2006). Analysis of family 1 UGT sequences have revealed that the N-terminal exhibits greater variability than the C-terminal, suggesting that amino acid residues in the former region bind to acceptors, while those in the latter accept donor substrates (Bowles et al., 2006, 2005).

Crystallographic analysis of mutant and wild-type UGTs from *Vitis vinifera* (VvGT1) and *Medicago truncatula* (UGT71G1) have provided considerable evidence clarifying the specificity and mechanisms of UGT reactions (He et al., 2006; Offen et al., 2006; Shao et al., 2005). A highly conserved histidine (His-20 in VvGT1 and His-22 in UGT71G1) in the PSPG box appeared to act as a crucial catalytic residue that triggers the hydroxyl group of glucosyl acceptors to assist in glucosidic linkage formation. Additionally, studies have identified a well-conserved aspartic acid residue (Asp-119 in VvGT1 and Asp-121 in UGT71G1) linked via hydrogen bond with the His residue, contributing to acid/base characteristics of the UGT throughout catalysis. Conversely, soybean UGTs exhibit catalytic mechanisms that do not involve a PSPG-box histidine residue, which they lack (Noguchi et al., 2007). More recently, two soybean UGTs have been functionally characterized as being involved in soyasapogenol G and R glycosylation (Shibuya et al., 2010).

In *Arabidopsis thaliana*, *Oryza sativa*, *Zea mays*, and *Cicer arietinum*, over 100 UGTs have been identified using the PSPG box (e.g., 120A. *thaliana* UGTs (Gachon et al., 2005; Osmani et al., 2009) and have been clustered into 14–17 major phylogenetic groups (A–Q) (Cao et al., 2008; Li et al., 2014, 2001; Sharma et al., 2014; Yonekura-Sakakibara and Hanada, 2011). Both lower (i.e., *Chlamydomonas reinhardtii* and *Physcomitrella patens*) and higher (i.e., *V. vinifera* and *Sorghum bicolor*) plants possess UGTs. Moreover, in vitro analysis of purified enzymes confirms the strong involvement of UGTs in plant growth and development, given major glycosylated substrates such as cytokinins (Hou et al., 2004; Wang et al., 2011), auxins (Jin et al., 2013; Tognetti et al., 2010), brassinosteroids (Poppenberger et al., 2005), salicylic acid (Lim et al., 2002), and abscisic acid (Priest et al., 2006). Some specific examples of UGT function include considerable changes to leaf size, shape, shoot branching, height, root growth, flowering time, and yield in *Arabidopsis* after the disturbance of hormone levels through UGT modification. Furthermore, UGT85A5 overexpression in transgenic tobacco improved salt tolerance (Sun et al., 2013), while *Arabidopsis* UGT73B5 and UGT73B3 were found necessary for resistance to biotic stresses (Langlois-Meurinne et al., 2005). The importance of diverse stress-responsive UGT genes in plant defense was further documented via microarray analysis showing the role of small-molecule glycosylation (von Saint Paul et al., 2011).

The role of UGTs in fiber and seed development, specifically, has been examined using RNA-seq data in cotton and flax (*Linum usitatissimum* L.) only (Barvkar et al., 2012). Here, we have employed bioinformatics techniques to identify 149 putative UGTs in soybean, and then subjected them to genomic, as well as expression, analyses. We then further confirmed the publicly

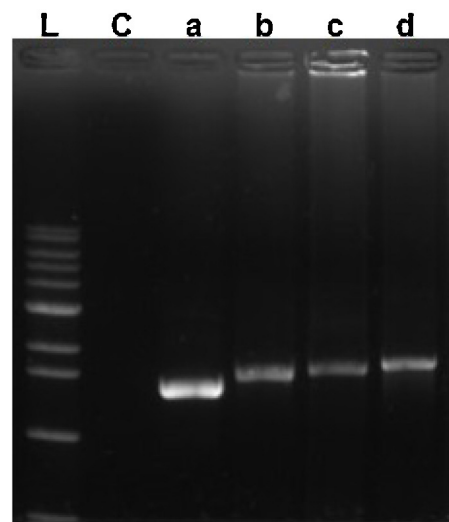


Fig. 1. Sequence confirmation through amplifying 4 UGTs from chromosome 10. (L) is the 1-kb ladder and (C) is control. (a) Glyma10g15790 (UGT98) 1386 bp. (b) Glyma10g07160.1 (UGT73) 1726 bp. (c) Glyma10g42680.1 (UGT73) 1742 bp. (d) Glyma10g15730.1 (UGT98) 1866 bp.

available microarray data on UGT expression during seed development with Affymetrix soybean IVT array and RNA-seq data. Finally, we designed UGT expression models to better understand expression patterns during seed development, using RPKM and clustering methods.

2. Materials and methods

2.1. Identification of UGTs in soybean

The draft genome of soybean (estimated size: 1115 MB with 46 430 genes) was downloaded from PlantGDB (<http://www.plantgdb.org/XGDB/phplib/download.php?GDB=Gm>) (Schmutz et al., 2010). Numerous plant databases were used to compose a dataset of 70 UGTs (Table S1) that generated the PSPG motif, using the PSWM (Position Specific Weight Matrix) tool. The predicted soybean proteome, with 64 675 sequences and 85 021 domains, was used to perform BLASTP (Basic Local Alignment Search Tool) 2.2.22 (Altschul et al., 1990), with the PSPG motif as a query. The expectation-value (E-value) cutoff was 1e-10. The PSPG motifs in all identified UGT proteins were further scanned with ExPASy ScanProsite (<http://prosite.expasy.org/prosite.html>). Their isoelectric point and molecular weight were calculated with ExPASy Compute pI/MW (http://web.expasy.org/compute_pi/). Phytozome (<https://phytozome.jgi.doe.gov/pz/portal.html>) was then used to obtain functional annotation for each UGT.

2.2. PCR amplification, cloning, and sequencing

To validate obtained UGT sequences, we extracted a genomic DNA from the 'Taekwang' soybean variety using the DNeasy Plant Mini Kit (Qiagen GmbH – Germany). Four UGTs were on chromosome 10 were randomly selected, and primers were designed to amplify the genes with their full reading frame, including start and stop codons (Table S2). The extracted DNA quality and quantity was measured using NanoDrop 1000 (Thermo Scientific Inc.USA). PCR amplification was performed using 50 ng of genomic DNA as template, with TAKARA Ex Taq HS polymerase (Clontech, Japan). Annealing temperatures are detailed in (Table S2). Targeted amplicons were visualized on 0.8% agarose gels and eluted with AccuPrep Gel Purification Kit (Bioneer Inc., Daejeon, Korea) (Fig. 1). Eluted

amplicons were cloned into pGEM-T Easy vectors (Promega, USA) and transformants were selected on ampicillin-resistance medium. The AccuPrep Plasmid isolation kit (Bioneer Inc.) was used to extract plasmids, and amplified products were sequenced via the BigDye™ terminator method in an ABI 3730XL DNA analyzer (Bioneer, Inc.).

2.3. Phylogenetic analysis of soybean UGTs

Multiple alignment of the identified UGTs in Clustal W with default gap penalties (<http://www.clustal.org/omega/>) (Thompson et al., 1994), and the phylogenetic tree was constructed with 100% coverage of the aligned data, using neighbor-joining (bootstrap value: 1000 replicates) in MEGA 6.0 (<http://www.megasoftware.net/>) (Huang et al., 2015; Li et al., 2014). Twenty-two reference UGT peptide sequences belonging to various phylogenetic groups (A–Q) were used for phylogenetic tree construction (Table S3). Total amino acid composition was also calculated using the MEGA sequence data explorer tool.

2.4. Intron and exon organization

A UGT intron map was constructed in accordance with a previously established method (Barvkar et al., 2012). Introns were classified based on structure in the genome, including phase, length, and number. The splice sites of intron-containing UGTs were mapped on all aligned sequences of intron-containing UGT peptides. Introns were sequentially numbered in relation to their positions in the amino acid sequence, created via aligning all genomic and coding sequences with Gene Structure Display Server (GSDS; <http://gsds.cbi.pku.edu.cn>) (Guo et al., 2007). Intron phases were determined as follows: phase 0 = introns positioned between two codons, phase 1 = introns positioned after the first base in the codon, and phase 2 = introns positioned after the second base in the codon. All intron sizes were recorded.

2.5. Chromosomal distribution and expansion

The physical location of each UGT on chromosomes was retrieved by using the soybean genome assembly Glyma.Wm82.a2 (Gmax2.0) at soybase.org. Map Chart 2.2 was used to visualize UGT distribution on soybean chromosomes (Voorrips, 2002). A gene cluster was defined as two or more copies located in a chromosomal region <200 kb (Holub, 2001).

To predict UGT expansion patterns, we looked for the occurrence of segmental and tandem duplication. Duplication for each UGT gene was obtained using the Plant Genome Duplication Database (PGDD) server (<http://chibba.agtec.uga.edu/duplication/>). Within a 100-kb range, anchors with identity >1.0 were rejected to avoid saturation (Yin et al., 2013). Assuming a molecular clock, the Ks values of duplicated genes are probably similar over time (Shiu et al., 2004); hence, to estimate dates of segmental-duplication events, Ks values were cast-off and their means calculated for each gene pair inside a duplicated block. The estimated date of the duplication event was then predicted with the mean Ks values ($T = Ks/2\lambda$), assuming clock-like rates (λ) of 1.5×10^{-8} identical substitutions/synonymous site/year for 6.1×10^{-9} (Lynch and Conery, 2000).

2.6. Detection of soybean UGT orthologs in five dicots

Orthologs of predicted UGTs were searched against Non-Redundant NCBI BLAST with the Blast2Go tool, setting an E-value of 0.001 (Sharma et al., 2014). A sequence similarity >80% was used for each UGT against five dicot genomic sequences: *P. vulgaris*, *V. angularis*, *V. radiata*, *P. montana*, and *M. truncatula*. These dicots were

selected based on genome homologies. Similarity percentages and the resultant accession numbers were recorded.

2.7. Digital expression analysis

Microarray data of putative UGTs were obtained from the Genevestigator database (<https://www.genevestigator.com/gv/>) against five developmental stages (flowering, fruit formation, germination, bean development, and main shoot growth) and 49 anatomical parts. A heat map of the data was generated following published methods (Li et al., 2014). This microarray data was further analyzed to clarify possible roles of UGTs during soybean seed development. First, the soybean mRNA profiling database tool (<http://seedgenenetwork.net/search?project=Soybean>) was used to generate a dataset of UGTs with differential expression, across 40 tissues/compartments, during four seed developmental stages: globular stage (GS), heart stage (HS), cotyledonary stage (CS), and early maturation stage (EMS). Data collection centered around probe set used, developmental stage, and tissues in which UGTs were expressed. The average signal intensity of each mRNA was captured. Signal intensities and detection calls (A, absent; INS, inconsistent detection call between biological replicates; M, marginal; P, present) were generated with the MAS 5.0 algorithm. For comparative purposes, Gene Chip data were scaled globally to a target intensity of 500 for all probe sets on the chip using MAS 5.0 default parameters. Each probe set was manually assigned a consensus detection call that was based on the program's detection calls for both biological replicates of RNA samples. Probe sets with identical calls across both biological replicates were assigned consensus detection calls of P, A, or M. By contrast, probe sets with discordant detection calls between the two biological replicates (e.g., P and A; P and M) were assigned a consensus detection call of Insufficient (INS). Images were captured from the website to visualize the data.

2.8. Transcriptome profiling and development of UGT models in soybean seed development

Digital expression of all UGTs related to seed development was further confirmed by RNA-sequence data. The 78 773 soybean gene models were filtered with the browsing tool of RNA-seq (<http://seedgenenetwork.net/search?project=RNA-Seq>), to retain those with RPKM values of zero to maximum in at least one of the four seed developmental stages (GS, HS, CS, and EMS). On average, one to six biological replicates were used for RNA-seq to obtain both RPKM values at each stage and tissues for each UGT. Normalized RPKM values were calculated for all UGTs, and then an average RPKM was calculated among all biological replicates. Maximum RPKM values were selected at each stage (irrespective of tissue), and UGT models were developed based on phylogenetic groups and UGT type.

3. Results

3.1. Identification of soybean UGTs

We identified 149 putative UGTs from the soybean projected proteome that possessed the PSPG motif at the C-terminal (Table S4 and Fig. S1). Lengths of the deduced UGT proteins varied from 345 to 587 amino acids, averaging at 472. The predicted molecular weight ranged from 39 to 60 kDa (Table S4). All UGT sequences started with methionine and were full-length. The isoelectric point ranged from 4.87 to 8.18. Of 149 UGTs, 65% were functionally annotated and exhibited different substrate specificities for anthocyanidin (2.4.1.297), soyasaponin (2.4.1.273), scopoletin (2.4.1.128), trans-zeatin (2.4.1.203), hydroquinone (2.4.1.218), isoflavone (2.4.1.170), 7-deoxyloganetin (2.4.1.324),

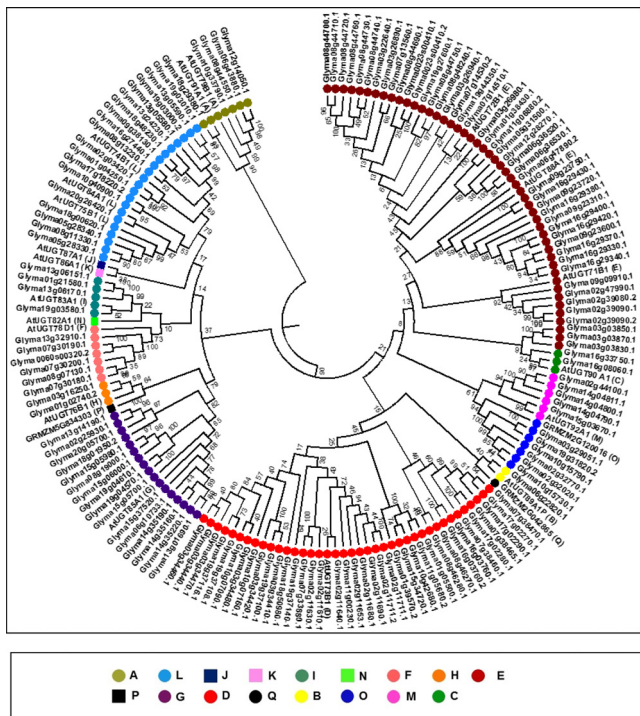


Fig. 2. Phylogenetic analysis of soybean UGTs indicated 15 distinct groups (A–O). The evolutionary history was inferred using the neighbor-joining method. The optimal tree with the sum of branch length = 2500.18896224 is shown. The percentages of replicate trees in which the associated taxa clustered together in the bootstrap test (1000 replicates) are shown next to the branches. Evolutionary distances were calculated as the number of amino-acid differences per sequence; the analysis involved 171 sequences. All positions containing gaps or missing data were eliminated, resulting in 106 genes total for the final dataset. Evolutionary analyses were conducted in MEGA6.

cyanohydrin (2.4.1.85), N-hydroxythioamide (2.4.1.195), and crocetin (2.4.1.271) (Table S4).

3.2. Phylogenetic analysis and amino acid composition of cultivated and wild soybean UGTs

Cloned UGTs from chromosome 10 were similar to sequences from the Plant GDB database. Multiple alignment of all sequences yielded a total of 781 aligned positions. Among them, 14 non-PSPG-motif positions were conserved, 659 were variable, and 44 were singletons. To validate our results, 128 UGTs from wild soybean (*Glycine soja*) were also identified with the same method (Fig. S1). These UGTs could be aligned in 1107 positions, with 15 non-PSPG conserved positions, 689 were variable, and 42 positions were singletons.

The constructed phylogenetic tree indicated a classification of 15 major groups (A–O), including a newly discovered group (O), which was absent in *Arabidopsis* but present in higher plants like maize and cotton (Fig. 2). The constructed cultivated soybean tree lacked the J, K, and N groups. To validate the deletion of these groups, we constructed another phylogenetic tree with wild soybean (*Glycine soja*) (Fig. S2). We found the same number of phylogenetic groups (A–Q) in wild soybean, but with C, J, K, and N groups absent. We also found one UGT sequence from wild soybean (A0A0B2QXX5) that did not cluster in any group and remained unclassified. Thus, J, K, and N groups were absent in both cultivated and wild soybean, whereas the C group was present only in cultivated soybean (Fig. 5). Sequences in the same group showed considerable similarities, ranging from 25% to 100%. Specifically, C and I groups shared 46%, B, G, M, and P groups shared 42%, and O

group shared 39% in its sequences. The lowest similarity percentage was found between E and D groups (25%).

Leucine was the most abundant amino acid across all UGTs from both cultivated and wild soybean (10%). Alanine, histidine, methionine, tryptophan, and tyrosine were the least common amino acids (1%–3%) (Fig. S3). All the amino acids were same in composition in wild and cultivated soybean except arginine, serine and valine (Fig. S3).

3.3. Intron/exon organization

Of 149 UGTs, 82 did not possess introns, 58 possessed one, and 7 possessed two (Table S5 and Fig. S4). A maximum of three introns were found in the following UGT genes: Glyma08g46280.1, Glyma14g04811.1, and Glyma0060s00320.2. A total of 68 introns were found across all intron-containing UGTs. Among them, seven introns were present in phase 2, 35 in phase 1, and 31 in phase 0 (Table S5 and Fig. S4). Collectively, intron sizes were in the range of 6–6299 bp. Of 68 introns, 9 were longer than 2000 bp (Fig. S5). A maximum of 12 introns were between 1 and 100 bp (Fig. S5). An average of 0.53 introns/gene was found among all intron-containing UGTs. Intron size was also compared between the phylogenetic groups: A (521–600 bp), D (19–6097 bp), E (42–961 bp), F (60 bp), G (147–5256 bp), I (726–1167 bp), L (168–1398 bp), M (512 bp), O (112 bp), and P (116–2347 bp). Group B and C did not have introns. An average of one to four exons was found among all UGTs (Table S4).

3.4. Chromosomal distribution and expansion pattern of the soybean UGTs

All soybean chromosomes except chromosome 04 had UGTs (Fig. 3), although the numbers varied. Chromosomes 02, 03, and 08 possessed the maximum number of UGTs (16, 15, and 17 respectively), whereas chromosomes 05, 12, and 20 had only two UGTs. A total of 58 clusters were found for all 149 UGTs, with sizes ranging from 2 to 12 genes per cluster (Fig. 3). Chromosomes 08, 03, 12, and 10 had the largest cluster. Maximum clustering was found between groups E and D. Twenty-five UGTs did not cluster.

Sixty-one UGTs out of 149 exhibited segmental duplication (Table S6). Out of these, 36% possessed two segmentally duplicated loci and 18% possessed three. Interestingly, two UGTs from group G, Glyma14g35160.1 and Glyma06g10733.1, were duplicated at five different loci. Moreover, tandem duplication was not observed. Phylogenetic groups E, D, G, and L possessed the maximum number of segmentally duplicated UGTs (13, 12, 12, and 9, respectively), while A and O possessed the four only. Group C and P did not have any segmentally duplicated UGTs.

The estimated time for segmental duplication in the 61 segmentally duplicated UGTs ranged from 6 to 182 mya. About 60% UGTs were segmentally duplicated 6–58 mya, while the remaining 40% occurred during 59–182 mya (Table S6). Ages of duplication were also calculated within each group (Table S6). Surprisingly, some UGTs with two different loci for duplication were duplicated at the same time but from different chromosomes. For example, Glyma03g34410.1 was segmentally duplicated simultaneously from Glyma.10G062200 and Glyma.19G187000 around 17 mya. Group F and G UGTs were commonly duplicated at three or four different loci across different times. For example, Glyma14g35160.1 and Glyma06g10733.1 were duplicated five times: in five different loci across five time periods. The same pattern was also observed in two UGTs from group F (Glyma07g30180.1 and Glyma08g07130), segmentally duplicated at three different loci within three separate time periods. Group O also showed the same pattern of duplication. Thus, soybean UGT evolution involves a series of segmental duplication, where members of groups A, B, D, E, I, L, and M were

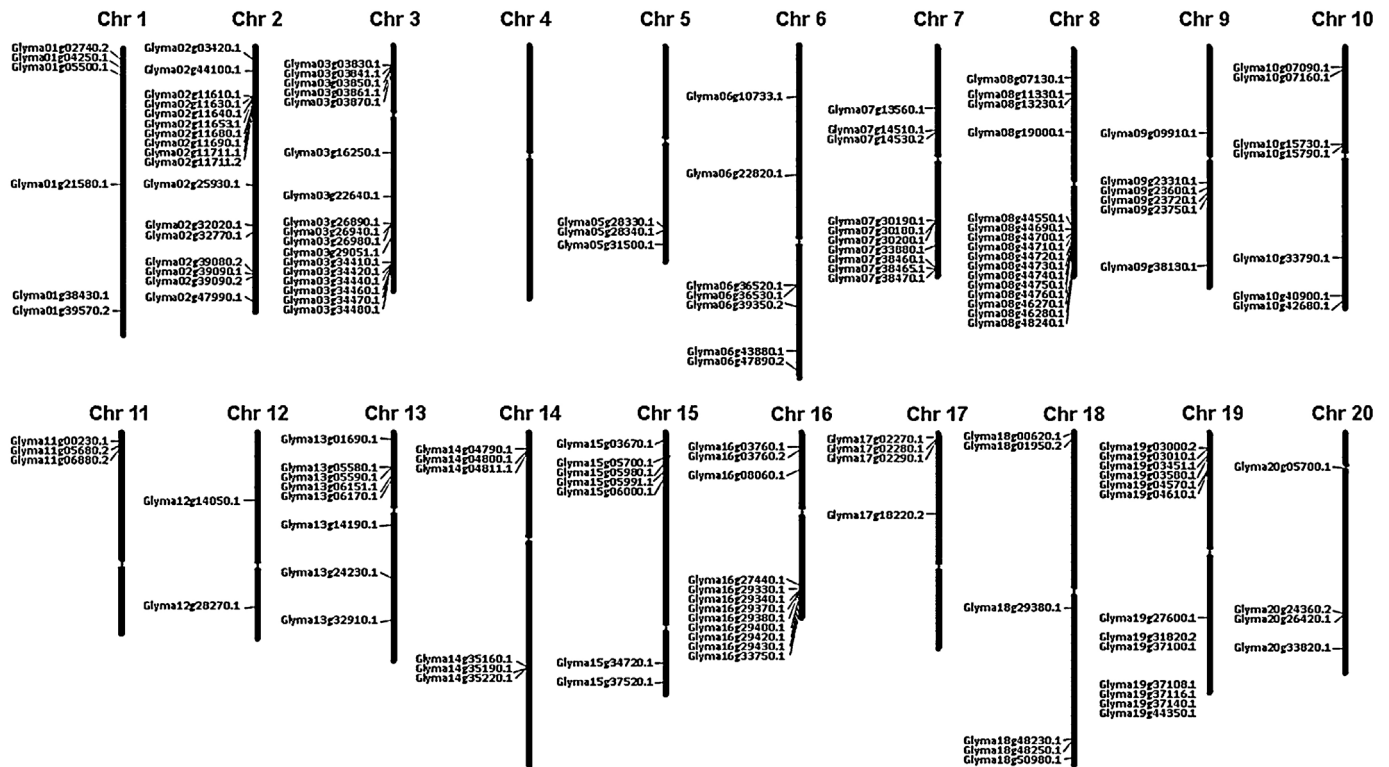


Fig. 3. Chromosomal distribution of 149 soybean UGTs across 20 chromosomes except chromosome 04. Clustering was based on this distribution.

duplicated twice, while members of groups D, F, and O members were duplicated three to five times. These segmental duplications may be independent and were incorporated into the soybean genome late.

3.5. Detection of soybean UGTs orthologs in other dicots

We identified 123 orthologs for 149 UGTs that were >80% similar. A maximum of 41 and 30 orthologs were detected from *P. vulgaris* and *V. angularis*, whereas *V. radiata* and *P. montana* both had 23 orthologs in soybean UGTs (Table S7 and Fig. S6). Phylogenetic groups E, D, L, and G possessed the maximum number of 34, 29, 22, and 15 orthologs across all five dicot genomes. In contrast, the O group had no orthologs. The maximum number of orthologs for group D came from *V. angularis* and *P. vulgaris*. Groups E, F, M, O, and I had no orthologs from *M. truncatula*. Some UGTs had orthologs from three different sources: Glyma16g27440.1 orthologs, for instance, were detected from *P. vulgaris*, *V. angularis*, and *V. radiata* (Table S7).

3.6. Digital expression of soybean UGTs

Microarray data from Genevestigator database identified 122 UGTs with varied expression across five developmental stages (Fig. S7A). Some of these genes were preferentially expressed at particular stages. For example, 60–70% expression was observed in these UGTs at bean development, germination, and main shoot growth stages, whereas 40–60% expression was observed during flowering and fruit formation stages. In addition, UGT expression was also observed across 49 anatomical parts (Fig. S7 B): high expression was found in testa, inflorescence, seed, cotyledons, meristems, shoot, mesophyll, and parenchyma tissues. Some UGTs were expressed in the root, pod, nodule seedling, shoot and leave tissues, and their corresponding cells.

To strengthen the possible role of UGTs during seed development, we found that 44 UGTs (30%) were actively involved during all four seed developmental stages (GS, HS, CS, EMS) and their respective tissues (Table S8). By using an Affymetrix soybean IVT array, we found that UGTs from groups A, C, D, E, G, I, L, and M contributed the most in soybean seed development. We also identified 56 publicly available probe sets to confirm the 44 UGTs expressed during seed development (Table S8). Glyma10g33790.1 and Glyma18g29380.1 from group A were expressed at three stages, but only in 11 and 9 tissues/compartments, respectively (Table S8 and Fig. S8). Glyma16g08060.1 from group C was only expressed in EMS and CS. Six UGTs from group D were expressed across all developmental stages and in almost all tissues, while the remaining four UGTs were only expressed in the EMS, CS, and HR stages (Table S8 and Fig. S8). Groups such as E, I, L, and M exhibited average signal intensities ranging from 5000 to 10000, indicating high expression throughout all stages and tissues. Glyma06g47890.2 was the only UGT to be maximally expressed in all stages and all tissues (Table S8 and Fig. S8). Four UGTs from group G showed signal intensities greater than 500 to 5000 across EMS, CS, and HS, with Glyma14g35160 being the most highly expressed.

3.7. Transcriptome profiling and UGT models for seed development

The results of RNA-seq analysis revealed that all 44 UGTs (Table S9) were notably expressed in at least one stage of seed development. On the basis of UGT type and phylogenetic groups, we constructed five gene models of UGTs expression during four seed developmental stages (Fig. 4). Group C UGTs were maximally expressed at GS, EMS, and CS, compared with group A UGTs (Fig. 4A). Group D UGTs were noticeably expressed in the GS and EMS stages, but lowly expressed in HS and CS (Table S9 and Fig. 4B). Glyma02g11640.1 from group D possessed the highest expression across all stages and a high RPKM value (Table S9 and Fig. 4B). Fif-



Fig. 4. UGT models derived from RNA-seq data. Normalized RPKM for all 44 UGTs was recorded on the basis of expression at seed developmental stages (GS, HS, CS and EMS). Phylogeny-based models represent UGT expression at four seed developmental stages: groups A and C in (4A), D in (4B), E and F in (4C), G and I in (4D), as well as M and L in (4E).

ten UGTs from group E were very lowly expressed throughout every stage (Fig. 4C). Glyma16g29430.1 from group E was maximally expressed at all stages. Group F UGTs were expressed during every stage, but not at very high levels. The UGTs from group G were normally expressed at all stages with an average RPKM. Glyma13g14190.1 from group G exhibited no expression across all stages, whereas Glyma08g19000.1 exhibited maximum expression at EMS (Table S9 and Fig. 4D). Group L UGTs were also normally expressed at all stages, with an average RPKM 1 to 70. Some UGTs, such as Glyma16g27440.1 and Glyma18g00620.1 from group L, exhibited maximum expression at EMS. All UGTs from Group M were highly expressed at all developmental stages, with RPKM value ranging from 3 to 230 (Table S10 and Fig. 4E).

4. Discussion

4.1. Family 1 UGT is a diverse and large family

The plant UGT multigene family has been identified in several species, including the model plants *Arabidopsis* and rice

(Caputi et al., 2012; Yonekura-Sakakibara and Hanada, 2011), but only analyzed in detail across *Arabidopsis*, flax, and cotton (Barvkar et al., 2012; Huang et al., 2015; Paquette et al., 2003). To deepen our understanding about this family in soybean, we performed a genome-wide analysis on phylogenetic relationships, intron/exon organization, expansion patterns, chromosomal distribution, amino acid composition, substrate specificity, expression patterns, and especially, the role of UGTs in seed development. The results indicated that the UGT multigene family is extremely diverse and large. Despite being an economically important crop, only two UGTs have been identified experimentally in soybean. Advances in genome sequencing tools greatly aid researchers in examining previously intractable data (based on biochemical studies) to obtain valuable information on gene identification. We identified 149 putative UGTs from soybean (Table S4), representing 0.26% of the whole genome. This percentage is higher than maize (0.23%) and lower than either flax (0.29%) or *Arabidopsis* (0.44%) (Barvkar et al., 2012; Li et al., 2014). Similar approaches were used for UGT identification in chickpea, cotton, maize, and flax (Barvkar et al., 2012; Huang et al., 2015; Li et al., 2014; Sharma et al., 2014).

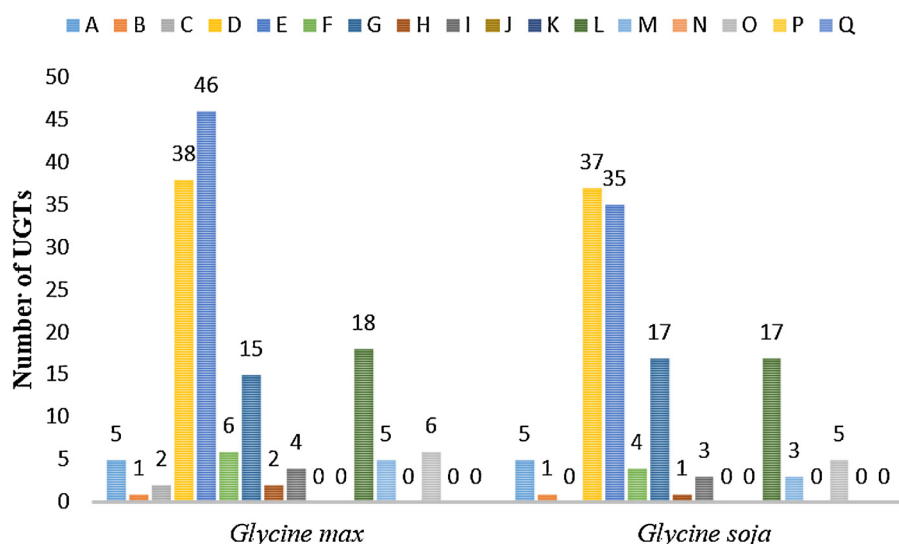


Fig. 5. Comparison of phylogenetic groups among all UGTs from cultivated and wild soybean. The E group in cultivated soybean was more extensive than the D, F, and L groups. The C group was absent in wild soybean, whereas J, K, and N were absent in both cultivated and wild soybean.

The abundance of UGTs across multiple plant genomes indicates their essential role in the glycosylation of numerous acceptor substrates with distinct functions.

4.2. Phylogeny, amino acid composition, and substrate specificity of soybean UGTs

Multiple sequence alignment of the putative soybean UGTs revealed extreme differences among their predicted protein sequences. Nevertheless, conserved domains in aligned amino acid sequences were clearly present (Fig. S1). This may not reflect the divergence in sequence length, but considerable variation was found among all UGTs: specifically, 64 to 84% positions were different from each other with respect to aligned positions in cultivated and wild soybean. The C-terminal region was far more conserved than the N-terminal region, confirming suggestions that the former is involved in binding the UDP moiety of the nucleotide sugar substrates (Li et al., 2001).

We identified 15 distinct phylogenetic groups (A–O) in cultivated and wild soybean, with an absence of J, K, and N groups in the former, as well as C, J, K, and N groups in the latter (Fig. 5). The same results were obtained in cotton, where group C was also absent in *Gossypium hirsutum* but present in *Gossypium arboreum* (Huang et al., 2015). These results suggest that the function of this group may be redundant with UGT members of another group. In this study, Group O was newly discovered, and can be found only in eight higher plant species (Caputi et al., 2012; Ross et al., 2001).

Our analysis showed that UGT79 and UGT91 enzymes cluster together, forming group A, while UGT92 sequences constitute group M, in agreement with previous work (Ross et al., 2001). Six UGTs were clustered into group O, supporting the view that members of this group can be found in higher plant species. Plant UGTs are thought to have undergone an expansion during evolution, with the development of new functions. Here, we observed that three phylogenetic groups (D, E, G) seem to have expanded more than any other groups (Fig. 5), an outcome generally conforming with an earlier study (Caputi et al., 2012), except for the case of group E. The number of putative genes in this group accounts for 31% and 27% in cultivated and wild soybean, respectively.

Indirect evidence exists indicating that amino acid composition of proteins depends on their dimension (Carugo, 2008). Leucine was the most abundant amino acid across all UGT peptide sequences, in both cultivated and wild soybean (Fig. S3). Here, we report for

the first time that UGTs are rich in leucine compared with other amino acids. The great degree of similarity between the wild and cultivated soybean UGTs suggest that this family remained highly conserved.

A major step forward in understanding UGT substrate specificity was the concept that the enzymes have regiospecificity or regioselectivity (Cartwright et al., 2008; Vogt and Jones, 2000). Phylogenetic analysis of plant UGTs showed that regiospecificity highly correlates with phylogeny; for example, group G members glycosylate cyanohydrin and 7-deoxyloganetin, while D, E, and L members glycosylate soyasapogenol B glucuronide, scopoletin, trans-zeatin O-beta, hydroquinone, anthocyanidin, isoflavone, N-hydroxythioamide S-beta, and crocetin. However, a key exception to this correlation is the development of substrate recognition via a region-switching event during evolution (Cartwright et al., 2008). As a result, AtUGT74F1 and AtUGT74F2 share more than 80% homology but use different substrates. AtUGT74F1 glycosylates the phenolic hydroxyl group of 2-hydroxy benzoic acid, while AtUGT74F2 glycosylates both the carboxyl and hydroxyl groups of 2-hydroxy benzoic acid.

4.3. Intron/exon organization in soybean UGTs

Our study revealed that about 58% of the UGTs contained introns (mostly one; Table S5 and Fig. S4). Comparing intron positions with sequence relationships predicted by the phylogenetic analysis, two conserved introns, C₁ and C₂, were identified (Fig. 6). They could be considered the oldest introns identified in UGT families. In soybean, C₁ and C₂ were present in most of the phylogenetic groups containing conserved introns (E, G, I, L, and M). This result suggests that both arose prior to UGT diversification. This hypothesis is also supported by the observation that most of the conserved introns were in the same phase. Nevertheless, conserved introns in the same phylogenetic groups maintained some differences, implying that the ancestor of this group contained only one conserved intron, and that intron loss and gain occurred after gene-duplication events giving rise to UGTs without introns or with non-conserved introns. In flax, two introns named I-3 and I-4 arose at the same position as what we found (Barvkar et al., 2012). In fact, the non-conserved introns exhibited a general pattern of gain during the evolution of the UGT gene.

Among the phylogenetic groups, intron phases remained consistent: groups A, C, G, I, and L possessed the same introns in the

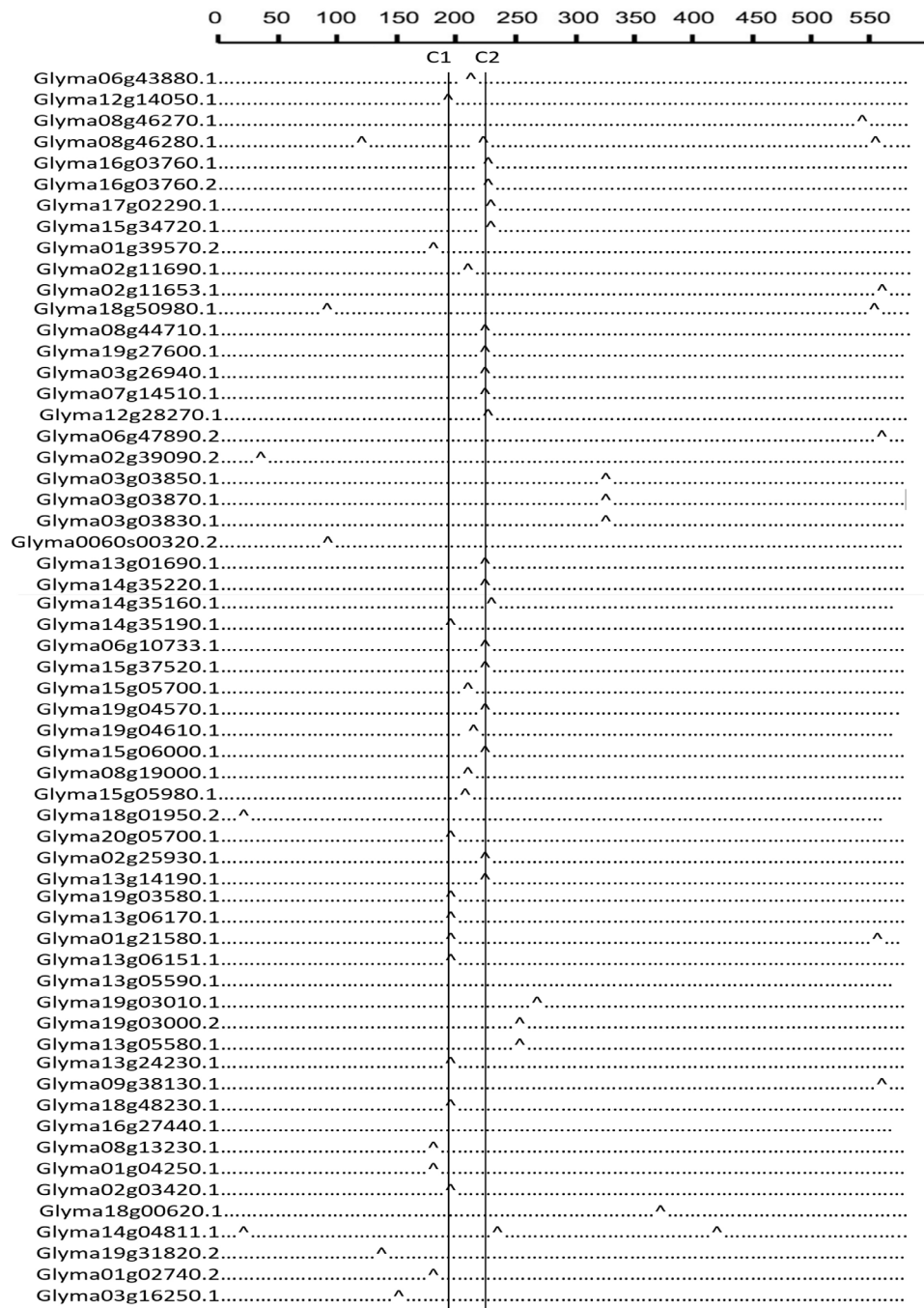


Fig. 6. Distribution of introns among intron-containing UGTs of cultivated soybean. The splice sites were marked relative to their positions on the amino acid sequence produced by multiple alignments of the intron-containing UGTs. Two conserved introns are denoted as C₁ and C₂.

same phase, which remained conserved within each group (Fig. S4). Group E has introns from all three phases, perhaps due to its wide expansion from other phylogenetic groups in soybean. Phase 1 had 57% introns and phase 0 had 51%, implying intron gain or loss within conserved periods during their evolution.

Intron size within each phylogenetic group seemed variable (Fig. S5). For example, Glyma08g46270.1 and Glyma15g34720.1 in group D possessed 85% similarity and the same intron phase, but differed in size (6097 bp and 424 bp introns, respectively) (Table S5). This suggests that intron size depended upon individual UGTs during evolution.

4.4. Chromosomal distribution and duplication of soybean UGTs

All UGTs were dispersed throughout the 20 chromosomes of the soybean genome except in chromosome 4 (Fig. 3). An obvious feature was that they were generally organized in clusters of 2 to 12 genes. Genes in the same cluster showed high-sequence similarity and were frequently classified into the same family or phylogenetic group. This tendency reflects their close phylogenetic relationships and the occurrence of recent gene duplication events. Similar observations were made in the *Arabidopsis* and cotton UGTs (Huang et al., 2015; Paquette et al., 2003; Ross et al., 2001), supporting the results presented here.

Sixty-one UGTs were segmentally duplicated within soybean genome (Table S6). The absence of tandem duplication in UGTs revealed that this family is evolved mainly through segmental duplication in soybean. These results are in the support of data showing that the soybean genome has undergone at least two rounds of genome duplication (Schmutz et al., 2010). Some UGTs from group G and F have evolved through a series of duplications, demonstrating that they have been individually duplicated and later became the part of genome.

Out of segmentally duplicated UGTs, 60% evolved during 6 to 58 mya, similar to previously estimated timing of genome duplication in soybean (Schmutz et al., 2010). The remaining 40% segmentally duplicated UGTs evolved 59 to 182 mya, again implying that these UGTs evolved independently and became part of soybean genome later.

The maximum number of orthologs from *P. vulgaris* supports the idea that the common bean is a diploid model for soybean (Fig. S6) (Schmutz et al., 2014). Phylogenetic groups G and L exhibited maximum orthology and similarity with common bean, suggesting that these groups evolved from the same origin but at different time periods. Some UGTs from D, E, L, M, and O groups were orthologs of *P. montana* (far relative of soybean) suggest that the soybean UGT family has a wide basis of evolution (Fig. S6).

4.5. Expression of soybean UGTs identified by microarray and supported by RNA-seq data

Until this study, the role of UGTs in seed development was not very defined, except in flax, and there is a considerable need for more exploration. It is known that flax seeds and stems contain a large number of secondary metabolites, possibly explaining the abundance of UGTs in these tissues (Kozłowska et al., 1983). This hypothesis was supported here by the results of the Affymetrix soybean IVT array and the finding that 44 UGTs from various phylogenetic groups (A, C, D, E, F, G, I, L and M) were actively involved across all four seed developmental stages, as well as in 40 tissues (Table S8 and Fig. S8). The active involvement of the above mentioned groups during seed development confirms their vital role of growth hormone glycosylation. The identified 56 probe set can be used to verify their expression at seed developmental stages and tissues. The recorded average signal intensity for each UGT was variable across tissues and stages. In each phylogenetic group, UGTs were variably expressed at specific stages and in specific tissues. For example, Glyma19g37140.1 and Glyma02g11640.1 from group D exhibited the lowest and highest signal intensities in seed development, suggesting that expression is based on individual UGTs even within a phylogenetic group. The maximum expression at EMS and GS by the D group UGTs suggests their active involvement for flavonol 7-O glycosylation because group D and B UGTs are annotated as flavonol 7-O-glucosyltransferase and 7-O rhamnosyltransferase, respectively (Jones et al., 2003; Yonekura-Sakakibara et al., 2007). Two UGTs from group D Glyma07g38460 and Glyma07g38470 have already been characterized for soyasaponin A glycosylation, and their presence was confirmed during seed development (Sayama et al., 2012). Group G UGT expression at EMS and CS confirmed their annotation for enhanced level of auxin biosynthesis.

Group L UGTs have been documented for indole-3-butyric acid glycosylation and their involvement in cotyledon development (Tognetti et al., 2010). We also found that Glyma18g00620.1 (UGT75L) was more highly expressed at HS, CS, and EMS, similar to results presented by (Zhang et al., 2016), indicating that UGT75D1 modulates cotyledon development in *Arabidopsis* (Fig. 4E). Group M UGTs possessed higher average signal intensities than any other group in all the seed stages; hence, they can be considered as potential targets for screening against these predicted classes to

identify their substrates. The maximum expression of group M UGTs during all stages confirmed their involvement in cytokinin glycosylation. Earlier studies postulated that UGT79, UGT91, and UGT92 enzymes from group A and M may act as cytokinin glycosyltransferase *in planta*, because their activity *in vitro* could not be demonstrated (Yonekura-Sakakibara and Hanada, 2011). Here, we also identified that Glyma06g47890 from group E was expressed in all stages and tissues during seed development, which suggests that this UGT can glycosylate several substrates. Group E includes UGTs with identified functions, namely abscisic acid glucosyltransferase (UGT71) (Priest et al., 2006) and monolignol 4-O-glucosyltransferase (UGT72, UGT88) (Lanot et al., 2006; Lim et al., 2001). Regarding unexpressed soybean UGTs, it is possible that some or most of these genes may be expressed at very low levels in specific tissues or are expressed only under certain conditions, such as biotic or abiotic stresses. Hence, they might have not been represented in microarray data, which was generated from unchallenged libraries. The expression analysis in this study gives us a global landscape of the expression of soybean UGTs in seed development.

All RNA-seq data were very similar to microarray and IVT data. For example, Glyma02g11640.1 from group D exhibited high average signal intensities and high RPKM values during seed development (Fig. 4 and Fig. S8). All 44 UGTs were maximally expressed at GS and EMS, suggesting that these two stages are highly important for glycosylation of growth hormones. In cotton and flax, UGT roles have also been confirmed in fiber and seed development, so our study corroborates earlier research (Barvkar et al., 2012; Huang et al., 2015). Overexpression of these UGTs in soybean can potentially improve several metabolic pathways that are necessary for plants to cope with environmental stresses, as well as increase the production of beneficial metabolites, such as saponins and isoflavones.

5. Conclusion

We identified 149 UGTs in the soybean genome. These genes were clustered into 15 distinct evolutionary groups (A–O) based on the phylogenetic analysis. One new UGT family member (O), not observed in *Arabidopsis*, was identified in the soybean genome. Two introns were gained prior to diversification in the soybean UGT family. Most identified genes were expressed in various tissue types, especially those related to during seed development, highlighting their deep involvement with this process. Results of the digital expression analysis were confirmed with RNA-seq data.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.jplph.2016.08.017>.

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